



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 3

Experiment Title:

**Design and Simulation of an 8-Bit Full Adder Using Verilog HDL
in Xilinx ISE**

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Date of Submission: [To be filled]

1 Title of the Experiment

Design, Synthesis, and Functional Simulation of an 8-Bit Full Adder Circuit Using Verilog HDL in Xilinx ISE Design Suite.

2 Objectives

The objectives of this experiment are:

- To design an 8-bit Full Adder using Verilog Hardware Description Language (HDL).
- To implement the arithmetic relationship between two 8-bit binary operands and a single carry-in bit.
- To generate and examine the RTL schematic of the designed circuit using Xilinx ISE.
- To observe the synthesized gate-level representation of the full-adder logic.
- To verify the functional correctness of the design using the ISim simulator with multiple input combinations.
- To observe the effect of the carry-in input and carry-out output during binary addition.

3 Theory

An 8-bit full adder performs binary addition on two 8-bit operands, $A[7 : 0]$ and $B[7 : 0]$, together with a single-bit carry-in, Cin .

The circuit produces an 8-bit sum, $Sum[7 : 0]$, and a single-bit carry-out, $Cout$.

The overall arithmetic operation can be expressed as:

$$\{Cout, Sum\} = A + B + Cin$$

For each individual bit position i , the full-adder equations are:

$$Sum_i = A_i \oplus B_i \oplus C_i$$

and

$$C_{i+1} = (A_i B_i) + (C_i (A_i \oplus B_i))$$

The eight bit positions operate together so that the carry generated at one bit position can propagate to the next higher bit.

4 Apparatus / Hardware & Software Requirements

- **Host PC:** Windows 10/64-bit environment
- **Virtualization Platform:** Oracle VM VirtualBox
- **Operating System Environment:** Oracle Linux 64-bit virtual machine
- **EDA Software:** Xilinx ISE Design Suite 14.7
- **Simulation Tool:** ISim
- **HDL:** Verilog HDL

5 RTL-Schematic Diagrams

The Xilinx ISE RTL viewer was used to inspect the structure of the 8-bit full-adder design after elaboration and synthesis.

5.1 Top-Level RTL Block Schematic

The top-level module, `full_adder_8bit`, accepts two 8-bit inputs, $A[7:0]$ and $B[7:0]$, together with the single-bit input Cin .

It produces the 8-bit output $Sum[7:0]$ and the carry-out $Cout$.

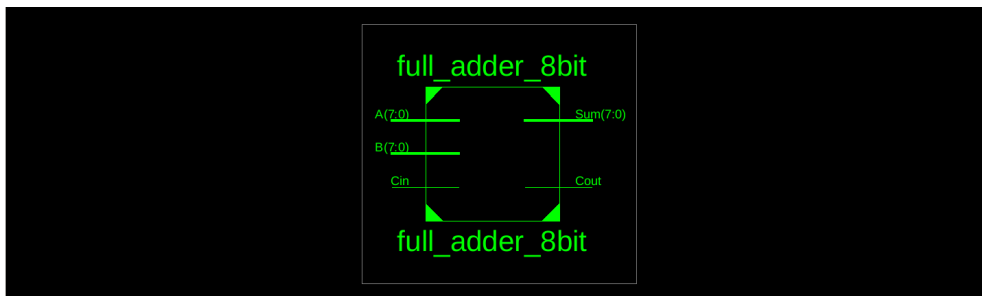


Figure 1: Top-level RTL block schematic of the 8-bit full adder.

5.2 Hierarchical RTL Schematic

The hierarchical RTL view shows the relationship between the `full_adder_8bit` top-level module and its internal `full_adder` logic block.

The 8-bit input and output buses are connected to the internal full-adder logic.

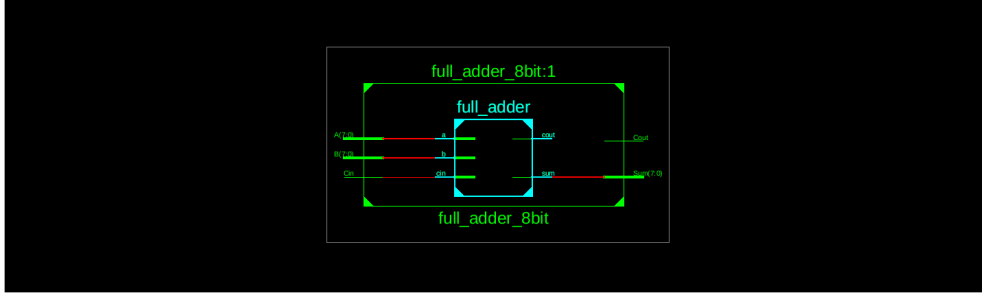


Figure 2: Hierarchical RTL schematic showing the internal full-adder module.

5.3 Synthesized Gate-Level RTL Schematic

The synthesized schematic expands the full-adder logic into basic combinational gates.

The displayed structure contains XOR gates for sum generation, AND gates for carry-generation terms, and an OR gate for combining the carry terms.

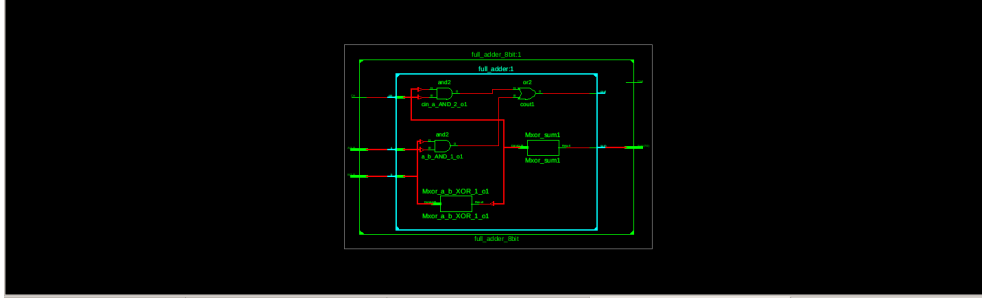


Figure 3: Synthesized gate-level RTL schematic of the full-adder logic.

6 Simulation

Functional simulation was performed using the ISim simulator in Xilinx ISE.

The testbench applies different combinations of the two 8-bit operands and the carry-in signal. The resulting $Sum[7:0]$ and $Cout$ signals are observed in the waveform viewer.

The captured simulation extends to approximately 220 ns and demonstrates both normal 8-bit addition and cases in which the result exceeds the 8-bit range, causing the carry-out signal to become high.

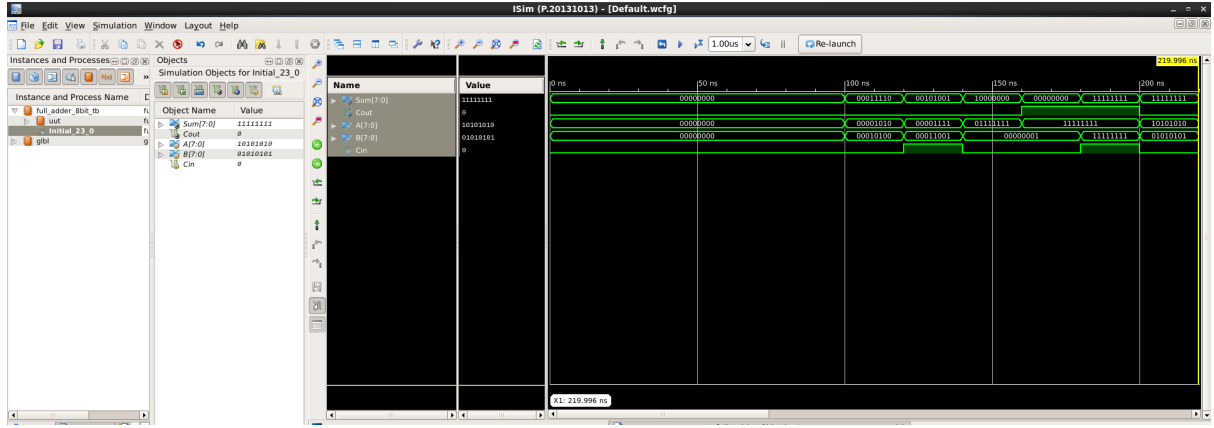


Figure 4: ISim functional simulation waveform of the 8-bit full adder.

6.1 Simulation Verification Table

The following table summarizes the visible test vectors in the simulation waveform.

The expected result is calculated from:

$$\{Cout, Sum\} = A + B + Cin$$

Table 1: Verification of simulated 8-bit full-adder outputs

Time	A	B	Cin	Expected $\{Cout, Sum\}$	Sim. $Cout$	Sim. Sum
0–100 ns	00000000	00000000	0	0 00000000	0	00000000
100–120 ns	00001010	00010100	0	0 00011110	0	00011110
120–140 ns	00001111	00011001	1	0 00101001	0	00101001
140–160 ns	01111111	00000001	0	0 10000000	0	10000000
160–180 ns	11111111	00000001	0	1 00000000	1	00000000
180–200 ns	11111111	11111111	1	1 11111111	1	11111111
200–220 ns	10101010	01010101	0	0 11111111	0	11111111

6.2 Selected Output Verification

The simulation demonstrates the following representative cases:

- $10 + 20 + 0 = 30$, giving $Sum = 00011110$ and $Cout = 0$.
- $15 + 25 + 1 = 41$, giving $Sum = 00101001$ and $Cout = 0$.
- $127 + 1 + 0 = 128$, giving $Sum = 10000000$ and $Cout = 0$.
- $255 + 1 + 0 = 256$, giving $Sum = 00000000$ and $Cout = 1$.
- $255 + 255 + 1 = 511$, giving $Sum = 11111111$ and $Cout = 1$.
- $170 + 85 + 0 = 255$, giving $Sum = 11111111$ and $Cout = 0$.

These cases verify both the lower 8-bit sum and the carry-out behavior of the circuit.

7 Code in GitHub Repository

The Verilog HDL source code and simulation testbench for this experiment will be maintained in a GitHub repository.

- **Repository:** <https://github.com/IstiaaNabil/Vlsi-Lab-Report>
- **Design Code:** Verilog HDL implementation of the 8-bit full adder.
- **Testbench:** Verilog HDL testbench used to generate the simulation input vectors.

8 Result

The 8-bit full-adder circuit was successfully designed in Verilog HDL and analyzed using Xilinx ISE.

The RTL viewer produced the expected hierarchical and gate-level representations of the combinational logic.

Functional simulation in ISim demonstrated correct 8-bit addition for the tested input combinations, including cases with and without carry-out.

9 Conclusion

In this experiment, an 8-bit Full Adder was designed and simulated using Verilog HDL in the Xilinx ISE Design Suite.

The RTL schematics were generated to examine the structure of the design, while the synthesized gate-level schematic showed the underlying XOR, AND, and OR logic used for full-adder operation.

The ISim waveform verified the arithmetic behavior of the circuit for several 8-bit input combinations.

The simulation also demonstrated correct carry generation when the arithmetic result exceeded the 8-bit range.

Therefore, the design, synthesis, RTL analysis, and functional simulation objectives of the experiment were successfully achieved.